

Demo

Doing logistic regression

Interpreting Logistic Regression

```
Call:
glm(formula = depress_dx ~ bp_dx, family = binomial, data =
hrs_df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	-1.8163	0.1112	-16.33	< 2e-16	***
bp_dx	0.5597	0.1258	4.45	8.61e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2341.2 on 2350 degrees of freedom
Residual deviance: 2319.9 on 2349 degrees of freedom
AIC: 2323.9

Number of Fisher Scoring iterations: 4

Dependent
Variable

Independent
Variable

```
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Each unit increase in `bp_dx` results in 0.5597 increase in the log odds of `depress_dx`.

```
data =  
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Null Hypothesis

There is no relationship

Alternative Hypothesis

There is a relationship

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Typically, we report the odds ratio (OR) which are the exponents of the coefficients.

Average increase in the odds of a depression diagnosis when there's a hypertension diagnosis.

Demo

Odds ratio

Demo

Multivariable Model

Knowledge Check

True or False: According to the model, age decreases the odds of depression diagnosis.

```
Call:
glm(formula = depress_dx ~ bp_dx + female + age_2014, family =
    binomial,
    data = hrs_df)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.983265	0.746024	1.318	0.188	
bp_dx	0.643142	0.129401	4.970	6.69e-07	***
female	0.874580	0.123961	7.055	1.72e-12	***
age_2014	-0.045736	0.009677	-4.726	2.29e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 2341.2 on 2350 degrees of freedom
Residual deviance: 2222.6 on 2347 degrees of freedom
AIC: 2230.6

Number of Fisher Scoring iterations: 4

**Respond on
Poll Everywhere**



<https://pollev.com/albertlee033>

Knowledge Check

True or False: According to the model, age decreases the odds of depression diagnosis.

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**Find out whether multivariable model
a better fit than the simple model
with the likelihood ratio test.**

Demo

Likelihood Ratio Test

Knowledge Check

True or False: Based on the results of this test, the multivariable model is a better fit.

Likelihood ratio test

Model 1: depress_dx ~ bp_dx

Model 2: depress_dx ~ bp_dx + female + age_2014

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	2	-1160.0			
2	4	-1111.3	2	97.355	< 2.2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

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Likelihood Ratio Test

Compare goodness of fit between two regression models if one is **nested**.

Full model: `depress_dx ~ bp_dx + female + age_2014`

Nested model: `depress_dx ~ bp_dx`

H_{null}: The full model and the nested model fit equally well.

H_{alternative}: The full model fits better than the nested model.

Knowledge Check

True or False: Based on the results of this test, the multivariable model is a better fit.

Likelihood ratio test

Model 1: depress_dx ~ bp_dx

Model 2: depress_dx ~ bp_dx + female + age_2014

	#Df	LogLik	Df	Chisq	Pr(>Chisq)
1	2	-1160.0			
2	4	-1111.3	2	97.355	< 2.2e-16 ***

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